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Version Migration as a Correlated Shock: Model Updates as an Unmanaged Risk Channel in LLM-Based Financial Systems

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ABSTRACT Financial institutions increasingly deploy large language models (LLMs) within credit and other decision-support workflows. However, vendor-driven model updates can alter behavior even when an institution holds its prompts, inputs, and downstream processes fixed. We quantify the decision instability associated with two prespecified model-version transitions: an OpenAI efficient-model update and a Google Gemini Flash update. Across both transitions, the experiment covers 100 companies, 12 analysis dates, and three replicates per version, producing 14,400 model calls. Identical inputs and matched per-cell seeds hold sampling conditions fixed, and within-version replicate disagreement provides an empirical noise floor. The primary endpoint is the excess cross-version flip rate above this floor. For the OpenAI migration, the cross-version disagreement rate exceeds the within-version noise floor by 11.2 percentage points (95% CI: 7.7–15.0); for the Gemini migration this excess is 38.1 percentage points (95% CI: 30.3–45.8). A prevalence–churn decomposition shows that the OpenAI update primarily reconsiders individual companies and shifts decisions toward greater severity, while the Gemini update produces a broad distributional shift toward the SAFE category. Neither migration shows clear improvement among observations whose labels change; the Gemini successor also shows lower aggregate class-balanced performance. Vendor version migration can therefore create material, directionally unpredictable decision discontinuities within otherwise unchanged LLM pipelines. We propose a pre-migration parallel-run control that measures excess flip rate, label-distribution shift, per-company churn, and directional drift before production cutover. The findings provide controlled evidence of deployment-level migration risk; broader cross-task and cross-institution effects remain for future study.

INDEX TERMS Algorithmic monoculture, credit risk, financial artificial intelligence, large language models, model drift, model risk management, model version migration, reproducible evaluation, third-party model risk.

I. INTRODUCTION

LARGE language models (LLMs) are increasingly incorporated into financial decision-support workflows [21], [22], including credit screening, counterparty assessment, disclosure review, and operational triage. In these applications, model outputs may influence downstream classifications, escalation decisions, or review priorities. Unlike conventional institution-developed models governed under established model-risk frameworks [7], [8], however, many LLMs are accessed through third-party application programming interfaces whose underlying model versions are controlled

by the provider. When an incumbent version is deprecated or replaced, institutions may be required to migrate even when their prompts, data, and downstream processes remain unchanged.

Such migrations create a distinct form of model-change risk. A successor version may preserve the behavior of its predecessor, in which case migration has limited operational consequence. Alternatively, differences in training data, architecture, optimization, or alignment may cause the same inputs to receive materially different decisions [1], [2], [19]. The size, direction, and structure of this discontinuity can-

not be inferred from the fact that a provider designates one version as the recommended replacement for another [24]. It must be measured empirically under controlled input and sampling conditions.

Existing evaluations of model updates face two measurement challenges. First, repeated calls to the same model version may produce different outputs because of residual sampling variability. A raw before-and-after disagreement rate may therefore attribute ordinary within-version variation to the migration itself. Second, an aggregate flip rate does not reveal how the change occurs. A successor may alter the overall distribution of labels across the evaluated portfolio, or it may reconsider individual companies while leaving the aggregate distribution approximately unchanged. These mechanisms have different operational and governance implications and should be measured separately.

This study evaluates two vendor-declared model migrations on a controlled financial classification task. Each incumbent and successor version receives identical, dated financial information and matched per-cell sampling seeds. Within-version replicate disagreement is used to estimate the ordinary noise floor, and the primary endpoint measures the cross-version flip rate in excess of that baseline. We further decompose observed migration changes into a marginal label-distribution shift and per-company churn, assess directional movement across credit-health categories, and compare both versions with a deterministic, rule-based Altman Z'' -based reference proxy [11], [12].

The paper focuses on deployment-level migration risk: the possibility that a provider-controlled update materially changes decisions within an otherwise fixed LLM pipeline. Because deployments using the same version encounter the same vendor-initiated transition, this mechanism can create synchronized changes across systems pinned to that version [3]–[6]. The study does not measure realized cross-institution or market-level effects; instead, it provides controlled evidence that the underlying decision-shock mechanism exists and can be quantified. In this study, “correlated shock” denotes a common upstream version change capable of affecting multiple deployments at approximately the same time; it does not denote an empirical estimate of cross-institutional correlation.

The principal contributions of this study are as follows:

- 1) **Controlled measurement of migration-induced decision instability:** We provide a preregistered evaluation of two real vendor migrations using an explicit within-version noise floor to distinguish update effects from ordinary sampling variability.
- 2) **Mechanism-level decomposition:** We separate migration-induced changes into marginal label-distribution shifts and per-company churn and report complementary chance-corrected agreement and directional-drift statistics.
- 3) **Outcome linkage:** We evaluate whether the decisions changed by each migration are associated with im-

proved classification quality against a deterministic, rule-based financial-health reference proxy.

- 4) **Governance application:** We translate the empirical measurements into a pre-migration parallel-run control that can detect material decision changes before production cutover.
- 5) **Reproducible research artifact:** We provide frozen inputs, exact model-version identifiers, matched seeds, raw responses, preregistration materials, and a deterministic analysis pipeline that regenerates the reported results.

II. RELATED WORK

A. MODEL-UPDATE REGRESSION IN GENERAL MACHINE LEARNING

Model updates can alter predictions even when task inputs and downstream decision rules remain unchanged. Prior work describes such instance-level regressions as negative flips, in which a successor model overturns a previously correct or stable prediction. MUSCLE examines this problem in LLM evolution and proposes compatibility mechanisms intended to reduce regressions between model versions [1], paralleling regression-free update objectives developed for vision models [26]. Related work on concept drift distinguishes genuine model changes from ordinary within-model variability [25], while standardized, holistic model-evaluation frameworks emphasize the need to evaluate successor versions before deployment [20]. Studies of task underspecification further show that apparently fixed prompts may leave substantial behavioral discretion to the underlying model [2], [27], allowing updates to change responses along dimensions not explicitly constrained by the prompt.

These studies primarily frame update regression as a model-performance, compatibility, or deployment-engineering problem. The present study examines a complementary question: how much a vendor-declared version migration changes categorical financial decisions after ordinary within-version response variability is removed. Rather than focusing only on whether individual predictions improve or deteriorate, we measure the aggregate decision discontinuity, its directional structure, and the mechanism through which the portfolio of classifications changes. This extends model-update research from instance-level compatibility to the operational risk created when a provider-controlled migration affects an otherwise unchanged financial decision pipeline.

B. ALGORITHMIC MONOCULTURE AND SYSTEMIC RISK IN FINANCE

A growing literature examines the possibility that widespread reliance on similar artificial intelligence systems may create correlated behavior across financial institutions and markets. Shared foundation models, common training data, and similar optimization objectives may cause otherwise independent users to produce aligned decisions, thereby reducing effective model diversity and increasing susceptibility to common failures [3], [23]. Related work has linked such convergence

to algorithmic herding, concentration risk, and broader forms of systemic fragility [4], [5], while supervisory research has begun to identify AI homogenization as a potential source of correlated financial risk [6], [28].

Most existing work addresses this concern conceptually or through market-level models of correlated trading and decision behavior. The empirical mechanism through which a common external change could simultaneously alter deployed decisions remains less directly measured. A vendor-controlled version migration provides one such mechanism: multiple deployments pinned to the same model version may encounter the same successor model and therefore experience a synchronized change in behavior. The present study evaluates this mechanism at the decision level by measuring how two vendor migrations alter fixed, graded credit-health classifications under identical inputs.

The analysis does not estimate realized systemic effects or cross-institution losses. Instead, the study establishes a mechanism through which deployments using the shared version could experience a common behavioral change. This decision-level evidence complements the systemic-risk literature by connecting abstract concerns about model concentration to a measurable operational event within an LLM-based financial workflow.

C. AGREEMENT, PREVALENCE, AND THE KAPPA PARADOX

Categorical agreement between an incumbent and successor model can be measured using chance-corrected statistics such as Cohen's κ [13], Scott's π [14], and Gwet's AC1 [10] (with Fleiss' generalization to multiple raters [15]). In this study, the two model versions are treated as paired raters that assign one of three credit-health labels (SAFE, WATCH, or DISTRESS) to the same company-date-replicate cells. These statistics summarize the extent to which the two versions produce consistent classifications beyond the level expected from their marginal label distributions.

Reliance on a single agreement statistic can be misleading when category prevalence is highly uneven or changes materially across versions. Cohen's κ and related measures may decline even when raw agreement remains substantial, a behavior commonly described as the κ paradox [16]. Gwet's AC1 is less sensitive to this prevalence effect, while Scott's π provides an additional chance-corrected comparison under a different expected-agreement formulation. Reporting all three statistics therefore allows the observed migration effect to be assessed for robustness to alternative treatments of marginal imbalance.

Agreement statistics alone, however, do not identify how a migration changes the portfolio of decisions. A low cross-version agreement value may result from a broad shift in label prevalence, from offsetting company-level reclassifications that leave the aggregate label mix largely unchanged, or from a combination of both. We therefore complement the agreement measures with an explicit prevalence–churn decomposition. The marginal component measures the minimum share of observations that must change to produce the

successor version's label distribution, while the remaining churn component captures company-level reshuffling that cancels in aggregate.

This combined framework separates three related but distinct quantities: overall cross-version consistency, distributional movement across credit-health categories, and genuine reclassification of individual companies. It therefore provides a more informative characterization of migration behavior than either raw flip rates or a single chance-corrected agreement statistic alone.

D. MODEL RISK MANAGEMENT AND GENERATIVE AI

Model-risk management frameworks emphasize validation, ongoing monitoring, change control, documentation, effective challenge, and institutional accountability for third-party models. In the United States, SR 11-7 historically established these expectations, but its lifecycle largely assumes that an institution can identify when a model changes and initiate review before the modified model enters production [7].

Vendor-hosted LLMs complicate this assumption because model versions may be updated, replaced, or deprecated according to the provider's schedule. A migration can therefore alter model behavior even when the institution holds its prompts, inputs, application logic, and downstream controls fixed. Recent research has proposed broader governance approaches for generative and agentic AI [9], but the present study addresses a narrower empirical problem: decision discontinuity caused specifically by vendor-directed model-version migration. We operationalize this risk through paired version testing, a within-version noise floor, and measures of excess disagreement, distributional shift, churn, and directional drift. The corresponding supervisory implications and proposed control are developed in Section VII.

E. REPRODUCIBILITY AND PREREGISTRATION

Empirical LLM evaluations are sensitive to model versions, prompts, sampling settings, data selection, exclusion rules, and statistical choices [31], while proprietary endpoint changes may prevent exact retrospective replay. We therefore specify the migration pairs, evaluation grid, prompt, seeds, endpoints, reference-label procedure, and decision rule before confirmatory data collection [30]; assign the frozen protocol a cryptographic hash; preserve immutable call-level responses and configurations; and regenerate all reported results through a seeded, build-validated analysis pipeline [32]. The archived responses, rather than future API replay, constitute the empirical record. Full procedural details are provided in Section IV-E and the Data and Code Availability statement.

III. PROBLEM FORMULATION AND METRICS

A. TASK AND UNIT OF ANALYSIS

For each vendor migration, let v_{old} denote the incumbent model version and v_{new} its designated successor. The unit of analysis is a matched company–date–replicate cell,

$$s = (j, t, r), \quad (1)$$

where j identifies the company, t the analysis date, and r the replicate. Each model version receives the same company information for the same cell and independently assigns one credit-health category,

$$y_{v,s} \in \{\text{SAFE}, \text{WATCH}, \text{DISTRESS}\}. \quad (2)$$

The model response is required to follow a fixed JSON schema containing the categorical label, a self-reported conviction score, and a free-text rationale. The categorical label is the output used in the present analysis.

The experimental comparison is paired. For every valid cell, the incumbent and successor versions are evaluated using identical financial inputs, prompt structure, inference settings, and matched cell-level seeds. Consequently, the model version is the principal experimental factor that varies between the two observations in each pair.

The primary estimand is not whether either version produces the correct financial classification. Instead, it is the extent to which replacing v_{old} with v_{new} changes the categorical decision produced by an otherwise unchanged pipeline. Reference labels are introduced separately for the secondary accuracy analysis and are not required to estimate migration-induced decision instability.

B. FLIP RATE AND THE WITHIN-VERSION NOISE FLOOR

A raw comparison between v_{old} and v_{new} may overstate migration-induced instability because repeated calls to the same model version can produce different labels under identical inputs. We therefore separate ordinary within-version variability from the additional disagreement introduced by the version change.

For version v , the within-version flip rate is defined as the proportion of replicate pairs for the same company and analysis date that return different categorical labels:

$$F_{\text{within},v} = \Pr(y_{v,j,t,r_1} \neq y_{v,j,t,r_2}), \quad r_1 \neq r_2. \quad (3)$$

Concretely, with three replicates per cell each cell contributes three unordered replicate pairs; $F_{\text{within},v}$ is the fraction of all such within-cell replicate pairs, pooled across cells, that disagree. The pooled within-version noise floor for a migration pair is the average replicate-disagreement rate across the incumbent and successor versions:

$$F_{\text{noise}} = \frac{1}{2} (F_{\text{within},v_{\text{old}}} + F_{\text{within},v_{\text{new}}}). \quad (4)$$

The cross-version flip rate is the proportion of matched cells for which the incumbent and successor versions return different labels:

$$F_{\text{cross}} = \Pr(y_{v_{\text{old}},s} \neq y_{v_{\text{new}},s}). \quad (5)$$

The preregistered primary endpoint is the excess flip rate,

$$F_{\text{excess}} = F_{\text{cross}} - F_{\text{noise}}. \quad (6)$$

This quantity is the cross-version disagreement rate in excess of the estimated within-version noise floor. It quantifies how much more the incumbent and successor versions disagree

than repeated invocations of a single version, and we interpret it as a noise-adjusted measure of cross-version disagreement rather than a formal identification of the causal share of decisions changed by the migration. A positive value indicates that the two versions disagree more than ordinary within-version sampling variability alone would produce.

Because the two versions may have different intrinsic variability, we also report a conservative robustness estimate that subtracts the larger of the two within-version floors:

$$F_{\text{excess}}^{\text{max}} = F_{\text{cross}} - \max(F_{\text{within},v_{\text{old}}}, F_{\text{within},v_{\text{new}}}). \quad (7)$$

The pooled estimator is used as the primary measure, while the maximum-floor specification serves as a stricter sensitivity analysis. The migration effect is considered supported when the estimated excess flip rate is positive and its 95% confidence interval excludes zero.

C. PREVALENCE-CHURN DECOMPOSITION

The cross-version flip rate measures the total proportion of decisions that change after migration, but it does not identify the structure of that change. A successor version may alter the overall distribution of SAFE, WATCH, and DISTRESS labels, or it may reclassify individual companies in offsetting directions while leaving the aggregate label shares approximately unchanged. We therefore decompose the gross cross-version flip rate into a marginal-distribution component and a company-level churn component.

Let $p_{\text{old}}(y)$ and $p_{\text{new}}(y)$ denote the proportions of observations assigned to category $y \in \{\text{SAFE}, \text{WATCH}, \text{DISTRESS}\}$ by the incumbent and successor versions, respectively. The marginal shift is defined as the total variation distance between the two label distributions:

$$M = \frac{1}{2} \sum_y |p_{\text{new}}(y) - p_{\text{old}}(y)|. \quad (8)$$

This quantity represents the minimum fraction of observations that must change labels to transform the incumbent version's aggregate label distribution into that of the successor version. It therefore captures portfolio-level re-rating attributable to a change in category prevalence.

Let F_{cross} denote the gross cross-version flip rate. The churn component is defined as

$$C = F_{\text{cross}} - M. \quad (9)$$

Churn captures additional company-level reclassifications that offset one another in the aggregate. For example, a movement from SAFE to WATCH for one company and a movement from WATCH to SAFE for another increase churn while leaving the overall category shares unchanged.

The decomposition therefore satisfies

$$F_{\text{cross}} = M + C, \quad (10)$$

where M measures distributional movement and C measures residual company-level reshuffling. A migration is described as marginal-dominated when $M > C$ and churn-dominated when $C > M$.

We complement this decomposition with three chance-corrected agreement statistics (Cohen's κ [13], Scott's π [14], and Gwet's AC1 [10]) computed between the paired incumbent and successor labels. Reporting multiple statistics reduces dependence on any single formulation of expected agreement, particularly when the category marginals are imbalanced.

As a supporting directional diagnostic, a continuity-corrected McNemar test [17] is applied to the binary indicator $\mathbb{I}(y = \text{DISTRESS})$, which examines whether movement into the most severe category differs from movement out of it. Because the McNemar test is applied at the matched-cell level without company-level clustering, its p -values are reported only as descriptive directional diagnostics and are not the primary inferential evidence for migration instability. This diagnostic supplements, and does not replace, the primary evidence, which follows the hierarchy: (i) the noise-adjusted excess cross-version disagreement; (ii) the company-cluster bootstrap confidence intervals; (iii) the prevalence–churn decomposition; (iv) the transition matrices; and (v) the McNemar test as a supporting directional diagnostic.

D. VERSION ISOLATION AND UNCERTAINTY QUANTIFICATION

To isolate the effect of model version, the incumbent and successor versions are evaluated on matched company–date–replicate cells using identical prompts, financial inputs, inference settings, and cell-level seeds. For each cell $s = (j, t, r)$, the seed is constructed as

$$\text{seed}_{j,t,r} = \text{SHA256}(\text{master} | j | t | r) \bmod 2^{31}. \quad (11)$$

The same seed is transmitted to v_{old} and v_{new} for the corresponding cell. This matched-seed design holds the nominal sampling condition fixed across versions, so that model version is the principal factor that differs within each comparison. The experimental logs verify that no seed mismatches occurred across the matched migration pairs. Because vendor seed support is best-effort rather than a guarantee of deterministic output, residual stochasticity is measured empirically through the within-version noise floor defined in Section III-B.

Only cells with valid categorical outputs from both the incumbent and successor versions are included in the paired cross-version analysis. This matched-pair requirement ensures that the flip rate, transition matrix, agreement statistics, and directional tests are computed from identical evaluation units. Call failures and invalid responses are reported separately so that any difference in matched sample size remains auditable.

Statistical uncertainty is estimated using a cluster bootstrap [18] with companies as the resampling unit. Clustering at the company level preserves dependence among repeated analysis dates and replicates for the same firm. For each of 2,000 bootstrap replications, companies are sampled with replacement, all associated dates, replicates, and paired version

outputs are retained, and the relevant statistics are recomputed.

Percentile-based 95% confidence intervals are generated using a fixed random-number-generator seed of 42. The pre-registered primary hypothesis is considered supported when

$$F_{\text{excess}} > 0 \quad (12)$$

and the corresponding 95% cluster-bootstrap confidence interval excludes zero. The same company-level bootstrap procedure is used for the conservative maximum-floor estimate and for the prevalence–churn quantities reported in the robustness analysis.

IV. EXPERIMENTAL DESIGN

A. EVALUATION GRID

The confirmatory experiment evaluates two vendor-declared model-version migrations over a common financial-classification grid. The grid contains 100 large-cap companies distributed across all 11 Global Industry Classification Standard sectors, twelve as-of dates in 2026, and three replicate calls for each company–date combination. The twelve as-of dates are 2026-01-09, 2026-01-23, 2026-02-06, 2026-02-20, 2026-03-06, 2026-03-20, 2026-04-03, 2026-04-17, 2026-05-01, 2026-05-15, 2026-05-22, and 2026-05-29; they span 9 January to 29 May 2026 at approximately two-week intervals, with the final three dates one week apart.

For each model version, the design therefore contains

$$100 \times 12 \times 3 = 3,600 \quad (13)$$

planned responses. Each migration pair compares an incumbent and successor version over the same grid, producing

$$3,600 \times 2 = 7,200 \quad (14)$$

planned responses per migration. Across the OpenAI and Gemini migration pairs, the complete confirmatory design contains 14,400 planned model responses.

The company universe is a fixed, prespecified list of 100 large-capitalization U.S.-listed companies, enumerated in the released study configuration and stratified across all 11 GICS sectors (between 8 and 10 companies per sector); the identical list is used for both migrations to support direct comparison of the magnitude and structure of version-induced changes. The set was curated to cover prominent large-capitalization constituents of each GICS sector rather than drawn programmatically from a named index on a fixed selection date; accordingly, no index-membership date or market-capitalization threshold is claimed. The exact constituents are released in full in the study configuration, so the universe is fully specified and reproducible even though it is not derived from a single external index. The analysis dates are also shared across versions within each migration pair. Each replicate is treated as a distinct model invocation with a matched seed across the incumbent and successor versions.

The grid is stratified by sector so that the estimated migration effect is not determined solely by one industry. Because

the Altman Z'' reference model is not designed for Financials and Real Estate, the primary migration-instability analysis includes all sectors, while the secondary accuracy analysis is additionally repeated after excluding those sectors.

B. NO-LOOKAHEAD FINANCIAL INPUTS

For each company–date cell, the model receives the most recent quarterly financial information that was publicly available before the corresponding analysis date. The applicable filing is selected using a strict no-lookahead rule: its filing date must precede the analysis date. No financial statement released on or after the evaluation date is included in the model input.

The filing data are converted into a standardized, human-readable summary containing the financial variables required for the credit-health assessment. These include revenue, profitability measures, assets, liabilities, working capital, retained earnings, and book equity. The same summary is supplied to both the incumbent and successor versions for every matched cell, ensuring that observed classification differences are attributable to model-version behavior rather than differences in the available information.

The input document does not disclose the Altman Z'' formula, its coefficients, or its classification thresholds. The models are instructed to make a judgment-style credit-health assessment from the reported financial information and return one of three labels,

$$\{\text{SAFE}, \text{WATCH}, \text{DISTRESS}\}. \quad (15)$$

The reference formula is applied independently after model inference and is used only in the secondary accuracy analysis.

All source financial data are retrieved once and frozen before confirmatory model evaluation. The reproducibility package records the selected filing date, source document, extracted financial fields, generated prompt content, and prompt hash for each company–date cell. This frozen-input procedure ensures that both versions are evaluated against identical evidence and that later changes in the underlying data source cannot alter the reported migration results.

C. RULE-BASED CREDIT-HEALTH REFERENCE

The secondary accuracy analysis uses a deterministic, rule-based credit-health reference proxy derived from the Altman Z'' model [12]. This reference is reproducible but is not treated as realized default ground truth; it serves only as a reproducible yardstick for the secondary accuracy analysis. For each company–date cell, the reference score is calculated from the same financial statements supplied to the LLMs, but the formula and thresholds are not shown to the models during inference.

The score is defined as

$$Z'' = 6.56X_1 + 3.26X_2 + 6.72X_3 + 1.05X_4, \quad (16)$$

where

$$\begin{aligned} X_1 &= \frac{\text{Working Capital}}{\text{Total Assets}}, \\ X_2 &= \frac{\text{Retained Earnings}}{\text{Total Assets}}, \\ X_3 &= \frac{\text{EBIT}}{\text{Total Assets}}, \\ X_4 &= \frac{\text{Book Value of Equity}}{\text{Total Liabilities}}. \end{aligned} \quad (17)$$

The resulting score is mapped to the three credit-health categories using the revised Z'' thresholds:

$$y_{j,t}^* = \begin{cases} \text{SAFE}, & Z''_{j,t} > 2.6, \\ \text{WATCH}, & 1.1 \leq Z''_{j,t} \leq 2.6, \\ \text{DISTRESS}, & Z''_{j,t} < 1.1. \end{cases} \quad (18)$$

These thresholds correspond to the revised non-manufacturer formulation rather than the original 1968 Altman Z -score [11]. The reference proxy is deterministic and reproducible for every company–date observation, allowing classification performance to be evaluated without relying on subjective analyst judgments, though it remains a rule-based proxy rather than realized default ground truth.

The rule-based reference is used only for the secondary accuracy analysis. The primary migration-instability measures (including the cross-version flip rate, within-version noise floor, excess flip rate, agreement statistics, prevalence shift, churn, and directional drift) are calculated entirely from the model outputs and do not depend on the reference label.

Because the Z'' model is not equally applicable across all industries, the full-sector analysis is supplemented by a robustness evaluation that excludes Financials and Real Estate. This distinction allows the migration effect to be measured over the complete company grid while limiting the interpretation of classification accuracy to sectors for which the reference model is more appropriate.

D. MODELS AND MIGRATION PAIRS

The study evaluates two vendor-declared model-version migrations, each involving an incumbent model and a vendor-positioned successor version. Exact snapshot identifiers are recorded and treated as experimental artifacts so that the analysis refers to specific deployed versions rather than broader model-family names, whose capabilities providers typically document only at the level of aggregate technical reports [29]. The evaluated migration pairs are summarized in Table 1; the exact model identifiers and each announced shutdown date are recorded from the providers' deprecation documentation as of the protocol-freeze date (4 July 2026) [33], [34]. The successor versions actually evaluated are fixed by the frozen experiment and are reported as run.

For each migration pair, the incumbent and successor versions are evaluated on the same company universe, analysis dates, financial inputs, prompt template, response schema,

TABLE 1. Evaluated migration pairs. Exact snapshot identifiers are pinned; the hash-frozen call-level dataset is the object of record.

Provider	Incumbent version	Successor version	Announced shutdown date
OpenAI	gpt-5-nano-2025-08-07	gpt-5.4-nano	December 11, 2026
Google	gemini-2.5-flash	gemini-3.5-flash	October 16, 2026

and matched cell-level seeds. No prompt modification, feature change, or downstream decision-rule adjustment is introduced between versions. This design isolates model version as the principal experimental factor.

The primary experiment is conducted at a decoding temperature of $T = 0$, which limits avoidable sampling variation while retaining any residual nondeterminism associated with the provider endpoint. Because zero-temperature inference does not guarantee identical outputs for proprietary hosted models, within-version replicate disagreement is estimated directly and incorporated into the primary excess-flip measure.

A separate robustness analysis evaluates both migration pairs at $T \in \{0, 0.7, 1.0\}$ on a smaller stratified subgrid. This analysis tests whether the sign of the migration effect is preserved under higher sampling variability. It is treated as supporting evidence rather than an independent replication of the full-grid confirmatory result.

The exact provider, model identifier, inference settings, seed, timestamp, prompt hash, token usage, and raw response are recorded for every attempted call. Because hosted model endpoints may change after the collection period, the frozen call-level dataset constitutes the empirical record of the evaluated migrations.

E. PREREGISTRATION, DATA FREEZING, AND REPRODUCIBILITY

The confirmatory design was specified before any model calls were made for the credit-health task. The frozen protocol defined the two migration pairs, company and date grid, prompt template, response schema, matched-seed construction, primary and secondary endpoints, reference-label procedure, exclusion rules, uncertainty-estimation method, and statistical decision rule. The complete protocol was assigned a SHA-256 hash to establish that these choices preceded confirmatory data collection.

The financial inputs, company universe, analysis dates, model-version identifiers, and experimental configuration were frozen before execution. Raw responses were collected once and stored in immutable, read-only, hash-stamped datasets. For each attempted call, the archive records the provider, exact model version, company, analysis date, replicate, matched seed, prompt hash, timestamp, token usage, response status, parsed label, and raw model output.

The analysis pipeline reads only the frozen datasets and configuration files. All stochastic procedures, including company-level bootstrap resampling, use fixed random-number-generator seeds. A build-validation step regenerates every reported statistic, table, and figure and rejects the

manuscript build if any reported value differs from the frozen analytical output.

Analyses defined before data collection are treated as confirmatory. Additional diagnostics or robustness checks introduced after the protocol freeze are identified separately as documented amendments and are not presented as part of the original confirmatory design. This separation preserves the distinction between prespecified hypothesis testing and subsequent exploratory analysis.

Because proprietary endpoints may not remain available or behaviorally identical after the study period, reproducibility is based primarily on the archived call-level responses rather than on the assumption that future API calls will reproduce the same outputs. The released package therefore includes the frozen inputs, raw responses, exact version strings, prompts, seeds, configuration files, analysis code, protocol hash, amendment log, and dataset manifests required to regenerate the reported results.

V. RESULTS

A. MIGRATION-INDUCED DECISION INSTABILITY

Both evaluated version migrations produce cross-version decision changes substantially above their respective within-version noise floors (Table 2, Fig. 1). The preregistered primary endpoint is therefore supported for both migration pairs.

For the OpenAI migration, 3,490 company–date–replicate cells contain valid, matched labels from both the incumbent and successor versions. The gross cross-version flip rate is 0.217, while the pooled within-version noise floor is 0.105. The resulting excess flip rate is

$$F_{\text{excess}}^{\text{OpenAI}} = 0.217 - 0.105 = 0.112, \quad (19)$$

with a 95% cluster-bootstrap confidence interval of [0.077, 0.150]. This represents an 11.2-percentage-point excess above the estimated within-version floor.

For the Gemini migration, all 3,600 planned cells contain valid, matched labels. The gross cross-version flip rate is 0.389, compared with a pooled within-version noise floor of 0.008. The excess flip rate is therefore

$$F_{\text{excess}}^{\text{Gemini}} = 0.389 - 0.008 = 0.381, \quad (20)$$

with a 95% cluster-bootstrap confidence interval of [0.303, 0.458]. For the Gemini migration, the cross-version disagreement rate therefore exceeds the within-version noise floor by 38.1 percentage points beyond ordinary within-version variation.

The difference between the two estimates is substantial. The Gemini migration produces an excess flip rate more than three times that of the OpenAI migration. However, both

TABLE 2. Primary and secondary results for both migrations (OpenAI 3,490 matched cells; Gemini 3,600).

Quantity	OpenAI	Gemini
Excess flip rate (primary)	0.112	0.381
95% CI	[.077,.150]	[.303,.458]
cross-version flip	0.217	0.389
within-version floor (pooled)	0.105	0.008
floor old / new	.120/.090	.016/.000
excess, max floor (95% CI)	.097 [.06,.13]	.372 [.30,.45]
Marginal (rating-mix) shift	0.095	0.319
Residual company-level churn	0.122	0.070
churn 95% CI	[.083,.161]	[.029,.120]
Cohen κ / Scott π / Gwet	.60/.60/.70	.33/.27/.47
McNemar drift (p)	6×10^{-18}	1.4×10^{-59}
Drift direction	DISTRESS	SAFE
Hit-rate old/new (chance .33)	.54/.56	.49/.43
Balanced acc. old/new	.54/.57	.49/.41
Macro-F1 old/new (baseline .36)	.52/.57	.46/.32
Flip-cond. accuracy Δ	+0.124	-0.179

confidence intervals exclude zero, demonstrating that neither result can be explained solely by residual sampling variability within the evaluated versions.

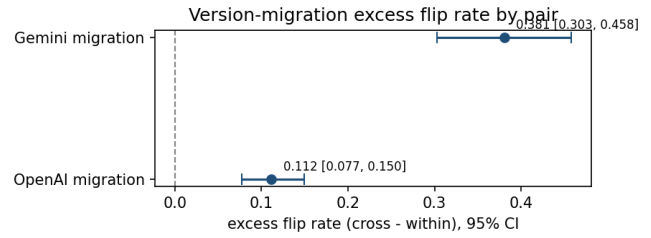
The matched-cell counts differ because the OpenAI endpoints did not return valid outputs for every planned call. Of 3,600 planned cells per version, the incumbent produced 3,540 valid labels and the successor produced 3,548. Requiring valid outputs from both versions at the same matched cell yielded 3,490 observations for the paired analysis. The missing observations resulted primarily from transient provider errors and one JSON-parsing failure and were distributed across sectors and dates. Gemini returned valid labels for all 3,600 cells per version after quota-limited calls were completed through a deferred backfill. All 3,600 Gemini successor responses passed the same schema and categorical-label validation applied to the other versions; no invalid, missing, or unparseable response was recoded as SAFE.

As a conservative robustness check, the excess flip rate is also calculated using the larger of the incumbent and successor within-version noise floors rather than their pooled value. Under this specification, the OpenAI excess flip rate is 0.097 with a 95% confidence interval of [0.060, 0.135], while the Gemini estimate is 0.372 with an interval of [0.295, 0.451]. Both remain positive with confidence intervals excluding zero. The primary finding is therefore robust to the method used to define the within-version noise floor.

B. STRUCTURAL DECOMPOSITION OF THE MIGRATION EFFECTS

Although both migrations produce statistically significant excess flip rates, the underlying structure of the decision changes differs materially across providers. The prevalence-churn decomposition separates the gross cross-version flip rate into a marginal label-distribution shift and residual company-level churn (Fig. 2).

For the OpenAI migration, the marginal shift is 0.095 and

**FIGURE 1.** Excess flip rate (cross-version minus within-version noise floor) with 95% CI for both migrations; both intervals exclude zero.

the churn component is 0.122, with a 95% confidence interval for churn of [0.083, 0.161]. Because churn exceeds the marginal shift, the OpenAI migration is primarily characterized by company-level reconsideration. A substantial share of firms move between categories in offsetting directions, so the aggregate label distribution changes less than the identities of the reclassified companies.

For the Gemini migration, the marginal shift is 0.319 and the churn component is 0.070, with a 95% confidence interval for churn of [0.029, 0.120]. Here, the marginal component is substantially larger than the churn component. The Gemini migration is therefore dominated by a portfolio-wide change in category prevalence rather than by offsetting company-specific reclassifications.

The difference can be summarized as

$$\text{OpenAI: } C > M, \quad \text{Gemini: } M \gg C, \quad (21)$$

where C denotes churn and M denotes marginal shift. The OpenAI successor primarily changes which companies receive particular credit-health labels, whereas the Gemini successor primarily changes the overall distribution of labels assigned across the evaluated portfolio.

Chance-corrected agreement statistics support the same distinction. For the OpenAI pair, Cohen's κ , Scott's π , and Gwet's AC1 are 0.60, 0.60, and 0.70, respectively. For the Gemini pair, the corresponding values are 0.33, 0.27, and 0.47. Although the absolute values differ across statistics because of their treatment of category prevalence, all three indicate materially greater cross-version consistency for the OpenAI migration than for the Gemini migration.

These results show that migration risk cannot be characterized by the excess flip rate alone. Two migrations may both produce material decision discontinuity while operating through different mechanisms: one through company-level reassessment and the other through broad movement in the portfolio's rating distribution.

C. DIRECTIONAL DRIFT ACROSS CREDIT-HEALTH CATEGORIES

The two migrations alter credit-health classifications in opposite directions (Table 3). The OpenAI successor shifts assessments toward more severe categories, whereas the Gemini successor shifts assessments toward the SAFE category.

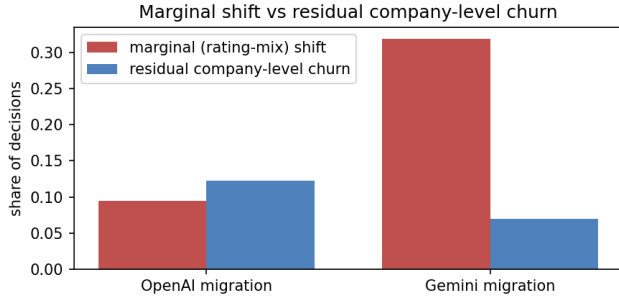


FIGURE 2. Prevalence decomposition of the gross flip into a marginal (rating-mix) shift and residual company-level churn. OpenAI is churn-dominated; Gemini is marginal-dominated.

For the OpenAI migration, the share of SAFE classifications decreases from 0.304 under the incumbent version to 0.209 under the successor. Over the same transition, the WATCH share increases from 0.588 to 0.636, and the DISTRESS share increases from 0.108 to 0.155:

$$\begin{aligned} p_{\text{SAFE}} &: 0.304 \rightarrow 0.209, \\ p_{\text{WATCH}} &: 0.588 \rightarrow 0.636, \\ p_{\text{DISTRESS}} &: 0.108 \rightarrow 0.155. \end{aligned} \quad (22)$$

The continuity-corrected McNemar test applied to the DISTRESS-versus-non-DISTRESS indicator yields

$$p \approx 6 \times 10^{-18}, \quad (23)$$

The McNemar diagnostic is consistent with asymmetric movement toward DISTRESS.

The Gemini migration exhibits the opposite pattern. The SAFE share increases from 0.389 to 0.708, while WATCH decreases from 0.525 to 0.281 and DISTRESS decreases from 0.086 to 0.012:

$$\begin{aligned} p_{\text{SAFE}} &: 0.389 \rightarrow 0.708, \\ p_{\text{WATCH}} &: 0.525 \rightarrow 0.281, \\ p_{\text{DISTRESS}} &: 0.086 \rightarrow 0.012. \end{aligned} \quad (24)$$

For this migration, the corresponding McNemar test gives

$$p \approx 1.4 \times 10^{-59}, \quad (25)$$

The diagnostic is consistent with asymmetric movement away from DISTRESS and toward less severe classifications.

The full 3×3 transition matrices reinforce these results (Fig. 3). For OpenAI, off-diagonal mass is concentrated toward more severe categories, while for Gemini it is concentrated toward SAFE.

These findings show that version migration does not introduce a common directional bias across providers. The OpenAI successor produces systematically more conservative credit-health assessments, whereas the Gemini successor produces systematically more lenient assessments on the same task and evaluation period. The direction of migration-induced drift is therefore provider- and version-specific rather than predictable from migration alone.

TABLE 3. Credit-label mix before/after each migration. OpenAI shifts toward DISTRESS; Gemini shifts sharply toward SAFE.

Version	SAFE	WATCH	DISTRESS
OpenAI v_{old}	0.304	0.588	0.108
OpenAI v_{new}	0.209	0.636	0.155
Gemini v_{old}	0.389	0.525	0.086
Gemini v_{new}	0.708	0.281	0.012

D. ROBUSTNESS OF CROSS-VERSION AGREEMENT

The comparison between the two migrations is robust to the choice of chance-corrected agreement statistic [10], [13], [14]. For OpenAI,

$$\kappa = 0.60, \quad \pi = 0.60, \quad \text{AC1} = 0.70, \quad (26)$$

whereas for Gemini,

$$\kappa = 0.33, \quad \pi = 0.27, \quad \text{AC1} = 0.47. \quad (27)$$

Although the absolute values differ because the statistics treat category prevalence differently, all three produce the same ordering:

$$\text{OpenAI agreement} > \text{Gemini agreement}. \quad (28)$$

The lower Gemini agreement is therefore not attributable solely to the prevalence sensitivity associated with Cohen's κ [16].

Migration effects are also larger among observations that are already unstable within the incumbent version. Relative to stable cells, the cross-version flip rate is higher by 0.330 for OpenAI and 0.250 for Gemini. This diagnostic pattern suggests that migration-induced changes are concentrated disproportionately near existing model decision boundaries.

E. SECONDARY ACCURACY ANALYSIS

The primary analysis measures migration-induced decision instability independently of any reference label. As a secondary analysis, we evaluate whether the successor versions improve classification quality against the deterministic, rule-based Altman Z'' -based reference categories. Because the three classes are imbalanced, performance is assessed using raw hit rate, balanced accuracy, and macro-F1 rather than raw accuracy alone.

For the OpenAI migration, classification performance changes as follows:

$$\begin{aligned} \text{Hit rate} &: 0.539 \rightarrow 0.564, \\ \text{Balanced accuracy} &: 0.538 \rightarrow 0.570, \\ \text{Macro-F1} &: 0.517 \rightarrow 0.565. \end{aligned} \quad (29)$$

For the Gemini migration, performance declines:

$$\begin{aligned} \text{Hit rate} &: 0.494 \rightarrow 0.425, \\ \text{Balanced accuracy} &: 0.488 \rightarrow 0.408, \\ \text{Macro-F1} &: 0.463 \rightarrow 0.325. \end{aligned} \quad (30)$$

The OpenAI successor produces modest numerical improvements, but the migration does not reliably improve accuracy

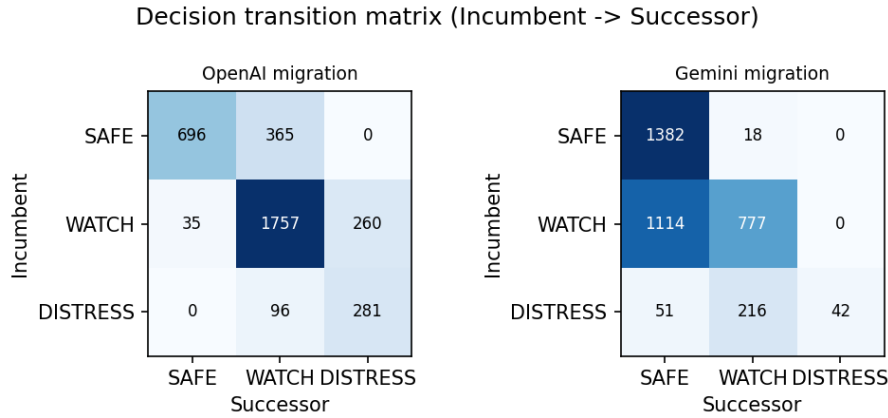


FIGURE 3. Decision transition matrices ($v_{old} \rightarrow v_{new}$) for each migration. OpenAI's mass moves off the diagonal toward more-severe tiers; Gemini's moves sharply into SAFE.

on the specific observations whose labels change. Conditional on a cross-version flip, the estimated accuracy change is $+0.124$, with a 95% confidence interval of $[-0.164, 0.418]$.

For Gemini, the successor's large shift toward SAFE is accompanied by lower class-balanced performance. The flip-conditional accuracy change is -0.179 , with a 95% confidence interval of $[-0.380, 0.030]$. The interval includes zero; the accompanying lower balanced accuracy and macro-F1 are consistent with the distributional shift providing no compensating classification improvement.

The sector-restricted analysis excludes Financials and Real Estate, for which the Altman Z'' reference model is less suitable. The principal pattern remains: raw hit rate changes from 0.590 to 0.562 for the OpenAI migration and from 0.553 to 0.491 for the Gemini migration.

These results show that substantial migration-induced decision changes do not necessarily correspond to improved classification quality. The OpenAI successor shows modest numerical improvement in aggregate metrics, whereas the Gemini successor shows lower class-balanced performance; neither migration shows clear improvement among the observations whose labels change. The accuracy analysis is secondary and does not affect the primary conclusion that both migrations produce cross-version disagreement well above the within-version noise floor.

F. TEMPERATURE SENSITIVITY

A supplementary temperature analysis evaluates whether the migration effect depends on the primary decoding setting of $T = 0$. Both migration pairs are re-estimated on a smaller stratified subgrid at $T \in \{0, 0.7, 1.0\}$.

For OpenAI, the estimated excess flip rates are 0.153, 0.167, and 0.208 at $T = 0, 0.7$, and 1.0, respectively (Table 4). For Gemini, the corresponding estimates are 0.222, 0.306, and 0.389. The estimated effect remains positive for both migration pairs at every tested temperature.

The confidence intervals are wide because the robustness grid contains only 24 company-date cells. In particular, the

TABLE 4. Temperature-sensitivity robustness (excess flip rate with 95% CI, 8×3 subgrid).

T	OpenAI	Gemini
0.0	0.153 [.042,.278]	0.222 [.000,.500]
0.7	0.167 [.056,.292]	0.306 [.111,.500]
1.0	0.208 [.069,.361]	0.389 [.167,.611]

lower bound for Gemini at $T = 0$ reaches zero. The temperature analysis should therefore be interpreted as evidence that the direction of the migration effect is preserved across the tested settings, not as an independently powered estimate of its magnitude.

The full-grid $T = 0$ analysis remains the primary result. The smaller temperature sweep provides supporting evidence that the observed version discontinuity is not limited to a single decoding temperature.

VI. DISCUSSION

Vendor-driven model-version migration is associated with materially different categorical financial decisions even when the prompt, inputs, inference settings, and downstream workflow remain fixed. After accounting for within-version variability, the noise-adjusted excess cross-version disagreement is approximately 11 percentage points for the OpenAI migration and approximately 38 percentage points for the Gemini migration. The two migrations also differ structurally: OpenAI is primarily churn-dominated and shifts assessments toward greater severity, whereas Gemini is primarily prevalence-dominated and shifts the portfolio toward SAFE. Migration behavior therefore cannot be assumed to be uniformly conservative, permissive, or performance-enhancing.

The prevalence-churn decomposition has direct operational significance. A prevalence-dominated migration changes the overall distribution of ratings and may affect portfolio limits, escalation volumes, and review capacity. A churn-dominated migration may leave the aggregate rating

mix comparatively stable while changing which individual companies are identified as risky. The secondary accuracy analysis further shows that decision turnover does not necessarily produce better classification quality. The OpenAI successor exhibits modest numerical improvement without reliable improvement among changed observations, while the Gemini successor exhibits lower balanced accuracy and macro-F1. Vendor migration should therefore be treated as a model change requiring validation rather than as an automatic upgrade.

The Altman Z'' -based label serves only as a deterministic reference proxy for the secondary outcome analysis [12]; the primary instability, agreement, churn, and directional-drift results depend solely on paired model outputs. More broadly, the findings establish a deployment-level mechanism through which systems using the same vendor version could experience a common behavioral change when migrated to the same successor [3], [6], [23]. The study does not measure cross-institutional adoption, realized losses, or market-level systemic effects. It demonstrates that the shared-update mechanism exists and can be measured, motivating the governance control developed in Section VII.

VII. GOVERNANCE: SR 11-7, SR 26-2, AND A PRE-MIGRATION CONTROL

SR 11-7 historically organized U.S. model-risk governance around independent validation, effective challenge, ongoing monitoring, change management, and institutional responsibility for third-party models [7]. Its lifecycle generally assumes that an institution can identify when a model changes and evaluate the modified model before production use. Vendor-managed foundation models weaken this assumption because the provider determines the successor version and deprecation schedule, while the surrounding prompt, application, and API interface may remain substantially unchanged. A migration can therefore substitute one decision function for another without appearing as a conventional redevelopment event.

The governance gap is sharper under the current framework. SR 26-2, which replaced SR 11-7 in April 2026, expressly excludes generative and agentic AI from its scope because of their novelty and rapid evolution [8]. The class of systems evaluated in this study therefore falls outside the revised guidance. Moreover, although SR 26-2 strengthens the treatment of concentration and dependency within an institution's model portfolio, it does not address correlated behavioral changes across institutions or deployments that depend on the same vendor version. The resulting gap is twofold: generative-AI systems are excluded from the revised framework, and the cross-deployment consequences of a common vendor update are not directly governed.

The empirical results connect to three established governance concerns summarized in Table 5. First, a vendor migration may constitute a material behavioral change even when no internal code or prompt changes. Second, limited deprecation notice may restrict the pre-production compar-

ison needed for effective challenge. Third, recording only the provider or model-family name is insufficient when exact snapshots exhibit materially different behavior. Exact version identifiers should therefore be maintained as governed model artifacts, and a version change should trigger formal testing, documentation, and approval.

Shared reliance on the same hosted version also creates a concentration channel beyond the model inventory of a single institution [6]. Multiple deployments may encounter the same provider-initiated change at approximately the same time. This study does not establish realized cross-institutional losses or systemic instability; it identifies the decision-level mechanism through which a common migration could produce correlated movement across deployments pinned to the shared version.

We propose a pre-migration parallel run conducted before production cutover. The institution applies the incumbent and successor versions to the same frozen, representative inputs using identical prompts, inference settings, and downstream decision rules. The control should estimate

$$F_{\text{excess}} = F_{\text{cross}} - F_{\text{noise}}, \quad (31)$$

together with the prevalence–churn decomposition and a directional-drift test. Where a defensible reference outcome exists, comparative performance should be evaluated as a secondary validation axis.

Cutover criteria should be specified before testing. Excess disagreement above an institution-defined materiality threshold, material directional drift, or disproportionate movement among high-risk cases should trigger expanded validation and documented approval rather than automatic deployment. Institutions should also inventory exact snapshot identifiers, seek advance deprecation notice, preserve access to the incumbent throughout the validation window, and retain frozen incumbent outputs as a challenger benchmark. The resulting parallel-run package provides evidence for monitoring, effective challenge, and change approval. This proposal is a practitioner-oriented control interpretation rather than authoritative supervisory guidance.

VIII. THREATS TO VALIDITY

Endpoint variability. Hosted LLM endpoints may remain stochastic even when temperature is set to zero and a seed is supplied. The study therefore does not assume deterministic model behavior. Instead, repeated within-version calls estimate residual variability, which is subtracted from the cross-version flip rate in the primary endpoint.

Reference-label limitations. Altman Z'' provides a deterministic, rule-based classification reference, but it is a proxy for realized credit outcomes and is not equally suitable for all sectors. It is therefore used only for the secondary accuracy analysis. The primary excess-flip, agreement, prevalence, churn, and directional-drift results do not depend on the reference labels.

Thin DISTRESS stratum. DISTRESS classifications constitute a relatively small share of the evaluated observations,

TABLE 5. Mapping vendor version migration to SR 11-7 provisions and the proposed control.

SR 11-7 provision	How version migration breaks it	Proposed control
Vendor/third-party responsibility & contingency planning	Anticipates vendor-model <i>loss</i> , not silent behavioral <i>replacement</i> on the vendor's schedule	Inventory the exact version string; pin the incumbent through validation
Ongoing monitoring; accelerate review on material model change	The change is exogenous and silent (API surface unchanged), so it may never trigger the review it warrants	Pre-migration parallel run detects the change and quantifies its materiality
Effective challenge; validation before use	Forced deprecation with limited notice removes the pre-change testable window	Parallel-run gate provides challenge and validation evidence before cutover
Model inventory, change control, documentation	Frozen documentation undermined by a model that changes underneath it	Treat version-as-artifact; hash-pin the snapshot as the object of record
Aggregate (institution-level) model risk	Not contemplated as correlated <i>across</i> institutions via a shared vendor update	Systemic gap; flagged for supervisory attention

so estimates involving movement into or out of the most severe category are based on fewer cells than the SAFE and WATCH comparisons. This does not affect the primary migration-instability result, which does not depend on the prevalence of the Altman reference label.

Task and prompt scope. The experiment evaluates one structured credit-health classification task using a single prompt family and standardized financial inputs. The measured effect sizes may differ for other prompts, domains, and input formats. The controlled design improves causal isolation and reproducibility but does not reproduce the full complexity of production workflows involving unstructured filings, retrieval systems, human review, or downstream automation.

Provider and migration-pair scope. Two migration pairs establish that material decision discontinuity can occur and that its direction and structure may differ across providers. They do not characterize the distribution of migration effects across the broader model ecosystem. Additional migrations and tasks are required before making claims about typical effect size, provider ranking, or model-family behavior.

Systemic-risk interpretation. The study measures synchronized decision changes at the deployment level, not realized cross-institutional losses or market instability. The results identify a mechanism through which deployments sharing the same model version could experience correlated movement. Whether this mechanism produces material system-wide effects depends on adoption concentration, institutional implementation, downstream controls, and economic exposure, none of which are observed here.

Single collection window. Confirmatory responses were collected once from exact model snapshots. Provider-side changes occurring during the collection period cannot be completely ruled out or averaged across repeated collection windows. The frozen call-level dataset is therefore the empirical object of record, and the findings apply to the evaluated snapshots and collection period.

IX. CONCLUSION

Vendor-managed LLM migrations are associated with material decision discontinuities within otherwise unchanged fi-

nancial workflows. After accounting for within-version variability, the noise-adjusted excess cross-version disagreement is 11.2 percentage points for the OpenAI migration and 38.1 percentage points for the Gemini migration. OpenAI primarily reclassifies individual companies and shifts decisions toward greater severity; Gemini primarily changes the portfolio-wide label distribution and shifts decisions toward SAFE. Neither migration shows clear improvement among observations whose labels change; the Gemini successor also shows lower aggregate class-balanced performance.

The study contributes a measurement framework that separates cross-version disagreement from ordinary endpoint variability and distinguishes distributional movement from company-level churn. These measures support a pre-migration parallel-run control through which institutions can evaluate successor models before production cutover.

The evidence is limited to two migration pairs, one structured classification task, and one prompt family. It does not estimate typical migration effects or realized market-level consequences. It nevertheless demonstrates a deployment-level mechanism through which systems using a shared vendor version may experience a common behavioral shock when that version is replaced.

Exact model versions should therefore be treated as governed artifacts, and vendor migrations should trigger documented testing, effective challenge, and change approval. The frozen, preregistered analytical package provides a reproducible basis for conducting and auditing such evaluations without assuming that future proprietary-endpoint calls will reproduce the archived responses.

DATA AND CODE AVAILABILITY

We release: (i) the immutable, read-only, SHA-256-stamped raw data (per-call model, seed, prompt hash, timestamp, tokens, cost, and raw response); (ii) the preregistration, its cryptographic freeze receipt, and the amendments log; (iii) the deterministic, seeded analysis pipeline that regenerates every table byte-for-byte, with a build gate that rejects any manuscript number disagreeing with the frozen analysis; and (iv) the capture and monitoring harness (the pre-migration parallel-run metric). **For review**, the complete artifact (code,

exact version strings, seeds, prompts, raw responses, and a manifest of SHA-256 hashes) is available anonymously at [anonymizedrepositoryURL], with the raw-response dataset deposited under an embargoed Zenodo record whose reviewer access token is provided in the submission system. The public Zenodo DOI will be released, and the repository de-anonymized, on acceptance; the frozen hashes let any reader verify the released data against the numbers reported here.

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REFERENCES

- [1] J. Zhang *et al.*, "MUSCLE: A Model Update Strategy for Compatible LLM Evolution," arXiv:2407.09435, 2024. [Online]. Available: <https://arxiv.org/abs/2407.09435>
- [2] C. Yang, Y. Shi, Q. Ma, M. X. Liu, C. Kästner, and T. Wu, "What Prompts Don't Say: Understanding and Managing Underspecification in LLM Prompts," arXiv:2505.13360, 2025. [Online]. Available: <https://arxiv.org/abs/2505.13360>
- [3] J. Kim *et al.*, "Correlated Errors in Large Language Models," arXiv:2506.07962, 2025. [Online]. Available: <https://arxiv.org/abs/2506.07962>
- [4] J. Danielsson, R. Macrae, and A. Uthemann, "Artificial intelligence and systemic risk," *Journal of Banking & Finance*, vol. 140, art. 106290, 2022, doi:10.1016/j.jbankfin.2021.106290.
- [5] R. Bommasani, K. A. Creel, A. Kumar, D. Jurafsky, and P. Liang, "Picking on the same person: Does algorithmic monoculture lead to outcome homogenization?" in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022. [Online]. Available: <https://arxiv.org/abs/2211.13972>
- [6] S. Cecchetti, R. L. Lumsdaine, T. Peltonen, and A. Sánchez Serrano, "Artificial Intelligence and Systemic Risk," Reports of the ESRB Advisory Scientific Committee, No. 16, Dec. 2025.
- [7] Board of Governors of the Federal Reserve System and OCC, "SR 11-7: Guidance on Model Risk Management," 2011.
- [8] Board of Governors of the Federal Reserve System, OCC, and FDIC, "SR 26-2: Revised Guidance on Model Risk Management," April 17, 2026 (supersedes SR 11-7); OCC Bulletin 2026-13.
- [9] National Institute of Standards and Technology, "Artificial Intelligence Risk Management Framework: Generative AI Profile," NIST Trustworthy and Responsible AI, NIST AI 600-1, Jul. 2024, doi:10.6028/NIST.AI.600-1.
- [10] K. L. Gwet, "Computing inter-rater reliability and its variance in the presence of high agreement," *British J. Math. Stat. Psychology*, vol. 61, pp. 29–48, 2008.
- [11] E. I. Altman, "Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy," *Journal of Finance*, vol. 23, no. 4, pp. 589–609, 1968.
- [12] E. I. Altman, "An Emerging Market Credit Scoring System for Corporate Bonds," *Emerging Markets Review*, vol. 6, no. 4, pp. 311–323, 2005 (see also Altman, Hartzell, and Peck, 1995), which defines the Z'' model and its non-manufacturer / emerging-market cutoffs (2.6 / 1.1).
- [13] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.
- [14] W. A. Scott, "Reliability of content analysis: The case of nominal scale coding," *Public Opinion Quarterly*, vol. 19, no. 3, pp. 321–325, 1955.
- [15] J. L. Fleiss, "Measuring nominal scale agreement among many raters," *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971.
- [16] A. R. Feinstein and D. V. Cicchetti, "High agreement but low kappa: I. The problems of two paradoxes," *Journal of Clinical Epidemiology*, vol. 43, no. 6, pp. 543–549, 1990.
- [17] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.
- [18] B. Efron, "Bootstrap methods: Another look at the jackknife," *Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.
- [19] R. Bommasani *et al.*, "On the Opportunities and Risks of Foundation Models," arXiv:2108.07258, 2021. [Online]. Available: <https://arxiv.org/abs/2108.07258>
- [20] P. Liang *et al.*, "Holistic Evaluation of Language Models," *Trans. Machine Learning Research*, 2023; arXiv:2211.09110.
- [21] S. Wu *et al.*, "BloombergGPT: A Large Language Model for Finance," arXiv:2303.17564, 2023. [Online]. Available: <https://arxiv.org/abs/2303.17564>
- [22] D. Araci, "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models," arXiv:1908.10063, 2019. [Online]. Available: <https://arxiv.org/abs/1908.10063>
- [23] J. Kleinberg and M. Raghavan, "Algorithmic monoculture and social welfare," *Proc. Nat. Acad. Sci.*, vol. 118, no. 22, e2018340118, 2021.
- [24] J. Wei *et al.*, "Emergent Abilities of Large Language Models," *Trans. Machine Learning Research*, 2022; arXiv:2206.07682.
- [25] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys*, vol. 46, no. 4, pp. 1–37, 2014.
- [26] S. Yan *et al.*, "Positive-congruent training: Towards regression-free model updates," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 14299–14308.
- [27] M. Sclar, Y. Choi, Y. Tsvetkov, and A. Suhr, "Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024; arXiv:2310.11324.
- [28] Financial Stability Board, "Artificial Intelligence and Machine Learning in Financial Services," Nov. 2017.
- [29] OpenAI, "GPT-4 Technical Report," arXiv:2303.08774, 2023. [Online]. Available: <https://arxiv.org/abs/2303.08774>
- [30] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proc. Nat. Acad. Sci.*, vol. 115, no. 11, pp. 2600–2606, 2018.
- [31] J. P. Simmons, L. D. Nelson, and U. Simonsohn, "False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant," *Psychological Science*, vol. 22, no. 11, pp. 1359–1366, 2011.
- [32] J. Pineau *et al.*, "Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program)," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.
- [33] OpenAI, "Deprecations," OpenAI API Documentation, 2026. [Online]. Available: <https://developers.openai.com/api/docs/deprecations> (accessed Jul. 4, 2026).
- [34] Google, "Gemini deprecations," Google AI for Developers, 2026. [Online]. Available: <https://ai.google.dev/gemini-api/docs/deprecations> (accessed Jul. 4, 2026).

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